

Lejun Min

Audio ML Research Engineer

- Stanford, CA
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- LinkedIn
- GitHub
- Mandarin / English

Profile

Audio ML researcher spanning generative audio and music, multimodal representation learning, and audio signal processing, from symbolic music generation to neural audio representations and deployed detection systems.

Technical Skills

Audio ML

- Audio & Music Generation
- Audio Tokenization / Neural Codecs
- Audio Representation Learning
- Audio Captioning
- Music Information Retrieval
- Audio Deepfake Detection
- Perceptual Audio Coding
- Digital Signal Processing

ML Methods

- Diffusion Models
- Transformers
- Self-Supervised Learning
- Multimodal Learning
- Contrastive Learning

Languages & Frameworks

- Python
- C++
- Rust
- PyTorch
- TensorFlow
- Hugging Face
- torchaudio
- librosa
- Accelerate
- Lightning

Audio Tools

- JUCE
- ChuckK
- Ambisonics

ML Systems

- Linux
- Docker
- Distributed Training
- SLURM
- MLflow

Education

Stanford University, CCRMA

2024 – Expected Jan. 2027

Master in Music, Science, and Technology (GPA 4.0/4.0)

- Advisors: Prof. Marina Bosi, Prof. Takako Fujioka
- Teaching Assistant of Perceptual Audio Coding (Winter 2026)

Shanghai Jiao Tong University, Zhiyuan College

2019 – 2023

Bachelor in Computer Science (Fellowship)

- Member of **ACM Honor Class** – elite CS program for top 5% students
- Top 1% Outstanding Bachelor's Thesis (41 out of 3873 graduates)

Experience

Qwen Audio, Alibaba Token Foundry – Research Intern

June 2026 – Sept. 2026

- Co-developed Qwen-Audio-3.0-Gen-Preview, a unified generative model for long-form, multi-character audio scenes with explicit temporal control.
- Improved downstream semantic alignment of an RVQ-based tokenizer by 25% using contrastive learning and multi-stage training.
- Contributed to curation and post-processing of 1M+ hours of audio data.

Smule Labs – Research Intern

Oct. 2025 – Dec. 2025, Mar. 2026 – May 2026

- Developed an end-to-end audio-captioning framework for disentangled music-semantic representations; achieved >0.4 Spearman audio-caption alignment in internal evaluations and validated specialization across five musical attributes.

Sony CSL Paris – Research Intern

June 2025 – Sept. 2025

- Built a two-stage generative music remixing system using mix-invariant representations and conditional generation.
- Reduced content-embedding variation across mixes of the same song by >90% in controlled ablations, without requiring stems at inference.

SoundPatrol & Stanford – Research Assistant

Feb. 2025 – May 2025

- Trained singer-identity representations for singing-voice deepfake detection; reduced EER by 10 percentage points and reached 96% top-1 accuracy on SingFake. The system has been integrated into SoundPatrol's infringement detection pipeline.

Music X Lab, MBZUAI – Research Assistant

June 2022 – Dec. 2022, Sept. 2023 – Feb. 2024

- Developed a four-stage cascaded diffusion system over a hierarchical symbolic language, generating structured pop songs up to 256 measures and significantly outperforming baselines on long-range structure (ICLR 2024 Spotlight, top 5%).
- Developed Polyffusion, a piano-roll DDPM supporting five generation and control tasks via masked inpainting and cross-attention; achieved the best objective results on 4/5 tasks and significant gains in naturalness and musicality (ISMIR 2023).

Selected Publications

Qwen-Audio-3.0-Gen-Preview Technical Report

arXiv 2026.07

J. Dai et al.; **L. Min** (all authors contributed equally)

Semantic Refinement of Universal Audio Representations

submitted to ICASSP 2027

L. Min et al.

Whole-song Hierarchical Generation Using Cascaded Diffusion Models

ICLR 2024 (Spotlight)

Z. Wang, **L. Min**, G. Xia [OpenReview] [Demo]

Polyffusion: A Diffusion Model for Polyphonic Score Generation

ISMIR 2023

L. Min, J. Jiang, G. Xia, J. Zhao [Poster] [Demo] [Code]

MuPT: A Generative Symbolic Music Pretrained Transformer

ICLR 2025

X. Qu et al.; **L. Min** (8/28) [OpenReview] [Demo]

Leveraging Rotational M/S Coding and ML in Stereo Audio Coding

IWSSPA 2025

L. Min, S. Chen, M. Bosi